

Brando Vasquez Rodriguez

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AVAILABLE 1st CO-OP SUMMER 2026 | 2nd CO-OP SPRING 2027

EDUCATION

Bachelor of Science, Electrical Engineering

University of Cincinnati, Cincinnati, OH

Graduating: May 2029

GPA: 3.97

High School Diploma

Fairfield High School, Fairfield, OH

May 2024

GPA: 4.3

SKILLS

Software: MATLAB, C++, Python, SQL, LabVIEW, Arduino IDE, LTspice, Verilog, Autodesk Fusion 360, HTML/CSS, Microsoft Office 365.

Hardware/Lab: Oscilloscope, Function Generator, Multimeter, Soldering, Embedded Systems, Sensor Integration.

Languages: English, Spanish.

PROFESSIONAL EXPERIENCE

Undergraduate Research Assistant

R.O.S.E Lab, University of Cincinnati

October 2025 - Present

- Analyzed time-series physiological datasets using HRV metrics and statistical methods to examine stress patterns in engineering students.
- Performed structured literature reviews and data organization to support experimental design and research methodology.
- Collaborate with faculty and interdisciplinary teams to maintain accurate research documentation and data reproducibility.

PROJECTS & ADDITIONAL EXPERIENCE

Light-Responsive Alarm System (ESP32 Project)

University of Cincinnati, Cincinnati, OH

March 2026 - May 2026

- Designed and implemented an ESP32-based embedded system integrating a phototransistor, temperature sensor, LCD display, and passive buzzer for real-time monitoring and responsive system behavior.
- Processed analog signals (12-bit ADC) and applied averaging to reduce noise and stabilize sensor measurements.
- Implemented Wi-Fi time synchronization and designed a custom CAD enclosure in Fusion 360 for the prototype system.

Student Ambassador

University of Cincinnati Blue Ash, Blue Ash, OH

August 2025 – May 2026

- Represented UCBA as a campus leader by serving as the first point of contact for over 4,800 students and their families.
- Conducted campus tours and orientations for approximately 1,800+ prospective students and guests.
- Served as a Student Life liaison, coordinating events and digital outreach with staff to enhance student engagement.

MATLAB Matrix & Graphing Tools (Project)

University of Cincinnati, Cincinnati, OH

August 2025 - November 2025

- Led a 3-person team developing MATLAB GUI tools for matrix computation, visualization, and numerical analysis.
- Implemented numerical algorithms, software logic, input validation, error handling, and dynamic plotting.
- Coordinated project planning, troubleshooting, documentation, and final presentation of the MATLAB-based application.

Python & SQL Trainee (Project-Based)

The Global Career Accelerator, Remote

May 2025 - August 2025

- Analyzed industry datasets for Intel and The Grammys to evaluate hardware sustainability and platform user engagement.
- Applied Python (Pandas, NumPy) and SQL to clean, transform, and analyze structured datasets in Jupyter Notebook.
- Created data visualizations and summarized findings into clear, actionable recommendations for stakeholders.

Autonomous Parking Robot (Project)

University of Cincinnati, Cincinnati, OH

January 2025 - April 2025

- Designed and built an autonomous LEGO EV3 vehicle using iterative engineering design and testing.
- Programmed multi-sensor control (color, gyro, ultrasonic, touch) in LabVIEW for autonomous parking.

LEADERSHIP, ACTIVITIES, & SERVICE

Vice President & Event Coordinator

Latinos en Acción (LEA) Affinity Club, University of Cincinnati Blue Ash

October 2024 - May 2025

- Led planning and execution of campus events, coordinating logistics, resources, and outreach to strengthen engagement.

ReStore Volunteer

Fairfield ReStore, Fairfield, OH

December 2024 - January 2025

- Volunteered 50 hours testing and troubleshooting electronics and appliances, supporting sustainable resale operations.